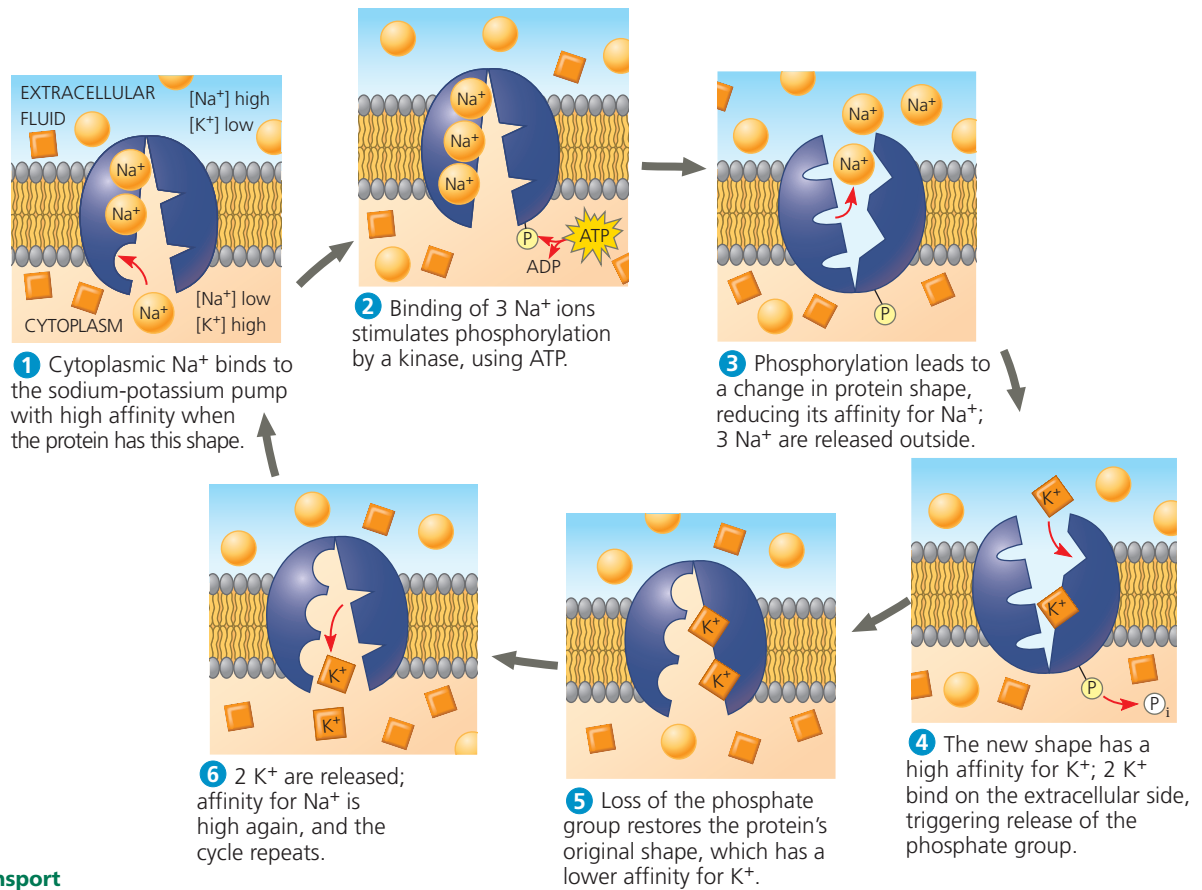


► **Figure 7.16 The sodium-potassium pump: a specific case of active transport.**

This transport system pumps ions against steep concentration gradients. The pump oscillates between two shapes in a cycle that moves Na^+ out of the cell (steps 1–3) and K^+ into the cell (steps 4–6). The two shapes have different binding affinities for Na^+ and K^+ . ATP hydrolysis powers the shape change by transferring a phosphate group to the transport protein (phosphorylating the protein).

VISUAL SKILLS For each ion (Na^+ and K^+), describe its concentration inside the cell relative to outside. How many Na^+ are moved out of the cell and how many K^+ moved in per cycle?

➔ **Mastering Biology**
Animation: Active Transport



an animal cell has a much higher concentration of potassium ions (K^+) and a much lower concentration of sodium ions (Na^+). The plasma membrane helps maintain these steep gradients by pumping Na^+ out of the cell and K^+ into the cell.

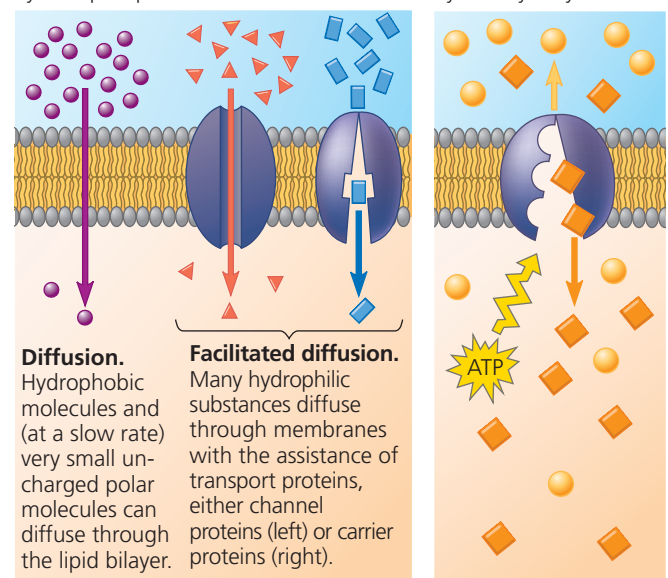
As in other types of cellular work, ATP hydrolysis supplies the energy for most active transport. One way ATP can power active transport is when its terminal phosphate group is transferred directly to the transport protein. This can induce the protein to change its shape in a manner that translocates a solute bound to the protein across the membrane. One transport system that works this way is the **sodium-potassium pump**, which exchanges Na^+ for K^+ across the plasma membrane of animal cells (**Figure 7.16**). The distinction between passive transport and active transport is reviewed in **Figure 7.17**.

How Ion Pumps Maintain Membrane Potential

All cells have voltages across their plasma membranes. Voltage is electrical potential energy (see Concept 2.2)—a separation of opposite charges. The cytoplasmic side of the membrane is negative in charge relative to the extracellular side because of an unequal distribution of anions and cations on the two sides. The voltage across a membrane, called a **membrane potential**, ranges from about -50 to -200 millivolts (mV). (The minus sign indicates that the inside of the cell is negative relative to the outside.)

▼ **Figure 7.17 Review: passive and active transport.**

Passive transport. Substances diffuse spontaneously down their concentration gradients, crossing a membrane with no expenditure of energy by the cell. The rate of diffusion can be greatly increased by transport proteins in the membrane.



VISUAL SKILLS For each solute in the right panel, describe its direction of movement, and state whether it is moving with or against its concentration gradient.